

Numerical Problems for Image Processing (With Solutions)

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Practice Numerical Problems with Solutions

Unit 1: Introduction — Problems and Solutions

1.1. Sampling & Quantization

- (a) A continuous image has spatial resolution 400 dpi. If sampled to 100 dpi, what is the reduction factor in total pixels?

Solution: Linear sampling factor = $\frac{100}{400} = \frac{1}{4}$. Area (total pixels) scales as square: $(1/4)^2 = 1/16$. The image has $1/16$ of the original pixels (reduced by factor 16).

- (b) An image of size 512×512 with 256 gray levels: (i) how many bits are required to store the image? (ii) How much memory (in KB) does it occupy?

Solution: Number of pixels = $512 \times 512 = 262\,144$. Bits needed = $262\,144 \times 8 = 2\,097\,152$ bits = $262\,144$ bytes = 256 KB.

1.2. Distance Measures

- (a) For pixels $p = (5, 5)$ and $q = (10, 13)$ compute Manhattan (D_4), Chessboard (D_8), and Euclidean distances.

Solution: $\Delta x = 5$, $\Delta y = 8$.

$$D_4 = |5| + |8| = 13.$$

$$D_8 = \max(5, 8) = 8.$$

$$D_E = \sqrt{5^2 + 8^2} = \sqrt{89} \approx 9.433.$$

Unit 2: Spatial Domain — Problems and Solutions

2.1. Gray-level transformations

- (a) Apply negative transformation with $L = 8$ to

$$\begin{pmatrix} 2 & 5 & 6 \\ 1 & 3 & 7 \\ 0 & 6 & 4 \end{pmatrix}.$$

Solution: Negative $s = 7 - r$ gives

$$\begin{pmatrix} 5 & 2 & 1 \\ 6 & 4 & 0 \\ 7 & 1 & 3 \end{pmatrix}.$$

(b) Apply log transform $s = c \ln(1 + r)$ (natural log, $c = 1$) to $r = 5, 10, 50, 200$.

Solution:

$$s(5) = \ln 6 \approx 1.7918, \quad s(10) = \ln 11 \approx 2.3979, \quad s(50) = \ln 51 \approx 3.9318, \quad s(200) = \ln 201 \approx 5.3033.$$

(c) **Histogram equalization**

(a) Given gray levels $\{0, 1, 2, 3\}$ with counts $\{100, 200, 300, 400\}$ perform histogram equalization (mapping, new histogram).

Solution: Total $N = 1000$. Probabilities $p = \{0.1, 0.2, 0.3, 0.4\}$. CDF $= \{0.1, 0.3, 0.6, 1.0\}$. Using mapping $s_k = \lfloor (L - 1) \cdot \text{CDF} \rfloor$ with $L = 4$ gives $(L - 1) \cdot \text{CDF} = \{0.3, 0.9, 1.8, 3.0\} \Rightarrow s = \{0, 0, 1, 3\}$. New histogram counts: level 0: $100 + 200 = 300$, level1: 300, level2: 0, level3: 400.

(d) **Normalized histogram**

(a) For the 4×4 image

$$\begin{array}{cccc} 0 & 1 & 1 & 2 \\ 2 & 2 & 3 & 3 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & 0 & 0 \end{array}$$

construct the normalized histogram.

Solution: Counts: $n_0 = 3, n_1 = 4, n_2 = 3, n_3 = 6$, total 16. Normalized histogram: $\{3/16, 4/16, 3/16, 6/16\}$.

2.2. Spatial filtering

(a) Apply 3×3 averaging filter to neighborhood $\begin{pmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \\ 1 & 2 & 3 \end{pmatrix}$.

Solution: Sum=45, average=45/9=5.

(b) Apply Sobel operator to

$$\begin{pmatrix} 10 & 10 & 10 \\ 0 & 0 & 0 \\ 10 & 10 & 10 \end{pmatrix}$$

Solution: Both G_x and G_y evaluate to 0; gradient magnitude 0 (no edge).

Unit 3: Frequency Domain — Problems and Solutions

3.1. DFT problems

- (a) Compute 1D DFT of sequence $\{1, 2, 3, 4\}$ ($N=4$).

Solution:

$$X(0) = 10, X(1) = -2 + 2i, X(2) = -2, X(3) = -2 - 2i.$$

- (b) Compute 2D DFT of $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ (2×2).

Solution: $F = \begin{pmatrix} 10 & -2 \\ -4 & 0 \end{pmatrix}.$

3.2. Filtering / FFT

- (a) For a 256×256 image with Ideal LPF cutoff $D_0 = 40$, state which components are retained.

Solution: All frequency indices (u, v) satisfying $\sqrt{(u - 128)^2 + (v - 128)^2} \leq 40$ (circle of radius 40 about center).

- (b) Gaussian LPF response for $D = 20, D_0 = 40$.

Solution: $H = e^{-D^2/(2D_0^2)} = e^{-400/3200} = e^{-0.125} \approx 0.8825.$

- (c) Show the first stage of FFT decomposition for sequence $\{1, 2, 3, 4, 5, 6, 7, 8\}$.

Solution: Even-indexed subsequence $\{1, 3, 5, 7\}$ and odd-indexed $\{2, 4, 6, 8\}$.

Unit 4: Restoration & Compression — Problems and Solutions

4.1. Noise models and filters

- (a) Gaussian noise: mean 20, variance 9. Produce five representative samples.

Solution: $\sigma = 3$. Representative samples (examples): $\{17, 22, 20, 25, 14\}$ (illustrative values near the mean).

- (b) Median filter on values $\{10, 20, 5, 3, 70\}$.

Solution: Sorted $\{3, 5, 10, 20, 70\}$, median = 10.

4.2. Mean filters

- (a) For the set $\{2, 3, 5, 10, 20\}$ compute: arithmetic, geometric, harmonic and contraharmonic ($Q=1$) means.

Solution: Arithmetic = 8.

Geometric = $(6000)^{1/5} \approx 5.695$.

Harmonic = $\frac{5}{1/2 + 1/3 + 1/5 + 1/10 + 1/20} = \frac{5}{1.183333} \approx 4.224$.

Contraharmonic (Q=1) = $\frac{\sum x^2}{\sum x} = \frac{538}{40} = 13.45$.

4.3. Compression numericals

- (a) Huffman coding for symbols $A(0.5), B(0.25), C(0.125), D(0.125)$.

Solution: Example code $A : 0, B : 10, C : 110, D : 111$. Average length = 1.75 bits/symbol. Compression ratio vs 2-bit fixed code = $2/1.75 \approx 1.1429$.

- (b) Run-length encode: 0, 0, 0, 255, 255, 128, 128, 128, 128, 0.

Solution: (0, 3), (255, 2), (128, 4), (0, 1).

- (c) 512×512 image (8 bpp) compressed to 1.5 bits/pixel: compressed size and ratio.

Solution: Pixels = 262 144. Compressed bits = $262\,144 \times 1.5 = 393\,216$ bits = 49 152 bytes = 48 KB. Original size = 256 KB. Compression ratio = $256/48 \approx 5.333$.

Unit 6: Segmentation — Problems and Solutions

6.1. Thresholding

- (a) Given values {10, 12, 14, 150, 160, 170} compute threshold using the iterative method.

Solution: Initial $T^{(0)} = (10+170)/2 = 90$. Group means $\mu_1 = 12, \mu_2 = 160$. New threshold $T^{(1)} = (12 + 160)/2 = 86$. Converges to 86.

6.2. Edge detection

- (a) Laplacian on

$$\begin{pmatrix} 5 & 5 & 5 \\ 5 & 10 & 5 \\ 5 & 5 & 5 \end{pmatrix}$$

with kernel $\begin{pmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{pmatrix}$.

Solution: Center output = $(-1)5 + (-1)5 + (-1)5 + (-1)5 + 4 \cdot 10 = -20 + 40 = 20$.

- (b) Roberts on $\begin{pmatrix} 5 & 8 \\ 6 & 10 \end{pmatrix}$.

Solution: $G_x = 5 - 10 = -5, G_y = 6 - 8 = -2, |G| = \sqrt{29} \approx 5.385$.

Unit 7: Representation, Description & Recognition — Problems and Solutions

7.1. Chain code

- (a) Boundary: $(1, 1) \rightarrow (1, 2) \rightarrow (2, 2) \rightarrow (3, 2)$. Give 4-direction code.
Solution: Using 4-direction ($0=E, 1=N, 2=W, 3=S$): moves are N,E,E $\rightarrow [1, 0, 0]$.

7.2. Fourier descriptors

- (a) Points $(1, 0), (0, 1), (-1, 0), (0, -1)$ as complex $z = [1, j, -1, -j]$. Compute 4-point DFT.
Solution: 4-point DFT yields $Z = [0, 4, 0, 0]$ (only first harmonic nonzero).

7.3. Perceptron training (one epoch)

- (a) Inputs and targets: $(0, 0) \rightarrow 0, (0, 1) \rightarrow 0, (1, 0) \rightarrow 0, (1, 1) \rightarrow 1$. Learning rate 1, initial weights 0. Perform one epoch.
Solution: With zero initial weights outputs are 0 for all inputs; only $(1, 1)$ has target 1 so update $w \leftarrow w + 1 \times 1 \times [1, 1] = [1, 1]$. Final weights after one epoch: $[1, 1]$.